INCH-POUND

MIL-B-49461/1C(ER)
13 August 1993
SUPERSEDING
MIL-B-49461/1B(ER)
10 March 1993

# MILITARY SPECIFICATION SHEET

## BATTERY, NON-RECHARGEABLE, LITHIUM THIONYL CHLORIDE, BA-6516/U

This specification is approved for use within Army Research Laboratory, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-49461(ER).

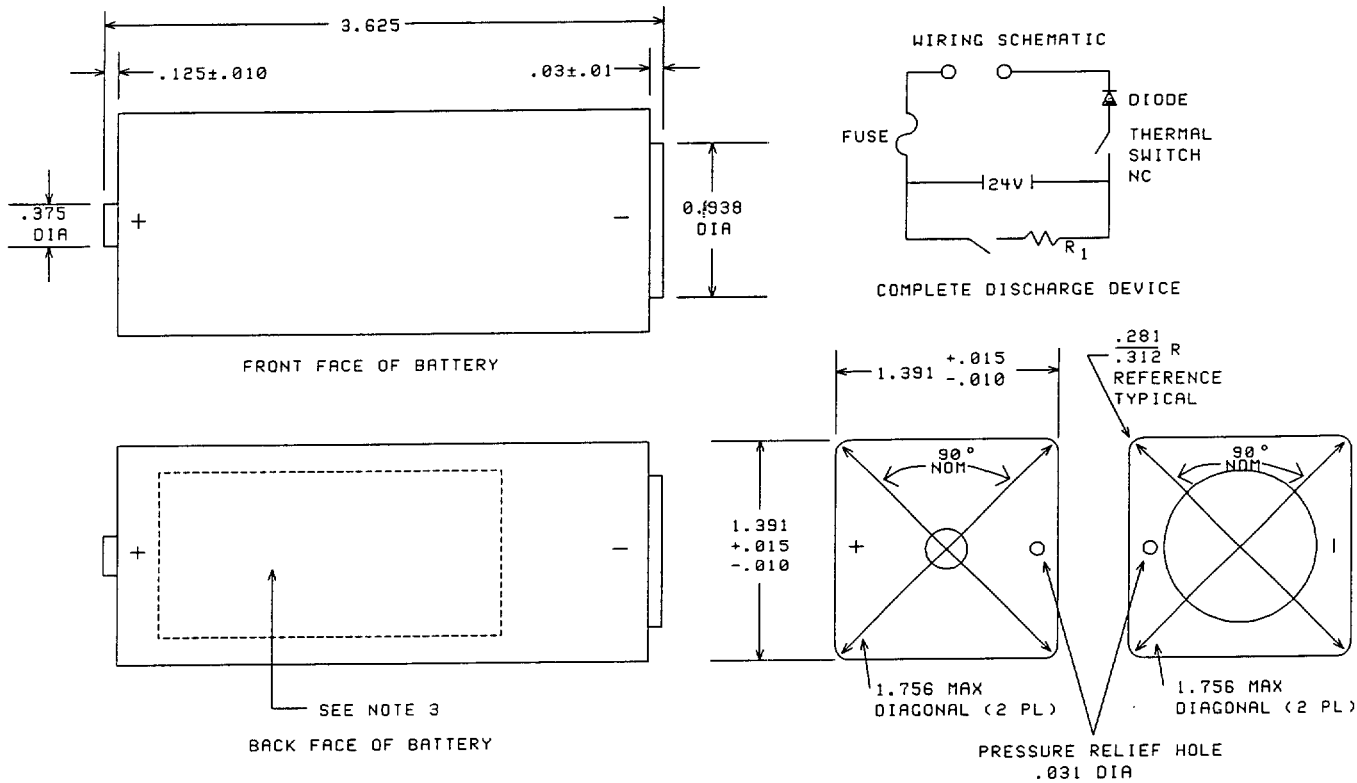



FIGURE 1. BATTERY DIMENSIONS AND LABELS.


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NOTES:

1. All dimensions are in inches.

2. Tolerances, unless otherwise specified, are ±.031 inch.

3. The identification label shall include the complete label information as required in 3.14 of the basic specification. If necessary, the label may be continued on an adjacent side.

4. The positive (+) terminal marking may be painted red. The negative (-) terminal marking may be painted black.

5. The diagonals for top and bottom views are 1.756 inches maximum. Diagonals shall not exceed 1.756 inch maximum along the length of the battery before or after testing or discharge.

6. $R_1$ value shall be between 900 and 1300 ohms and have a power rating of 1.0 watt.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

No load voltage: Shall not exceed 30 volts.

Nominal voltage: 24 volts.

End voltage: Cutoff voltage shall be 20 volts.

The battery shall contain a 2.0 ampere non-replaceable slow blow fuse in the negative leg.

The two terminals shall be low resistance flat contacts of nickel per MIL-N-46025.

Weight (maximum): 0.6 pound (272.4 grams).

Capacity tests:

(1) Storage: Details on storage conditions for all capacity tests are specified in the basic specification.

(2) Discharge: Batteries shall be pulse discharged at 1.5A for 1 second, then off for 9 seconds. This 1 second on, 9 seconds off cycle shall be repeated continuously until zero volts is reached.



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However, time to 20 volts shall be used for calculation of capacity and/or service life. For the I capacity test, the batteries shall be pulse discharged at reverse (negative) voltage for 0.7 amp-hours immediately after zero volts is obtained. There shall be no venting, fire, or explosion when the batteries are force discharged.

| Capacity Test | Battery Capacity in Amp-Hours to 20.00 volts |
|---|---|
| I | 1.1 |
| L | 0.5 |
| H | 1.05 |
| HT | 1.0 |
| LT | 0.3 |
| T | 1.05 |
| D | 1.05 |
| IT | 1.0 |

Initial voltage delay: When the battery is subjected to any of the capacity tests, the time required at the beginning of discharge for the battery to reach a voltage of 20.00 volts after the specified load is applied shall not be more than 1.0 minute.

High temperature switch: A high temperature switch or thermal fuse shall be employed internal to the battery as described in paragraphs 3.5.5 and 3.5.5.1 of the basic specification.

Complete discharge device: This device shall be inserted in the battery as shown on the circuit schematic and shall have the characteristics as described in paragraph 3.5.5.3 of the basic specification.

Flame test: Not applicable.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodian:
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(Project 6135-A262)



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